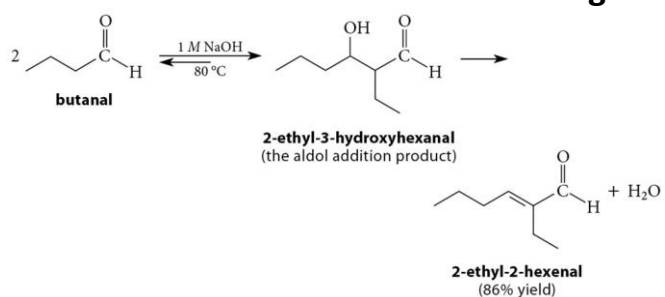
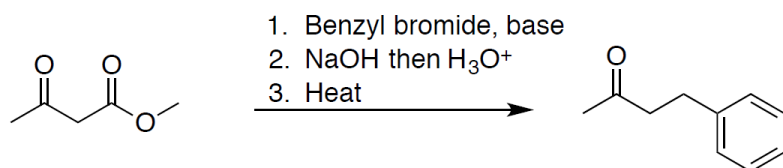


Provide a mechanism for the following reaction.

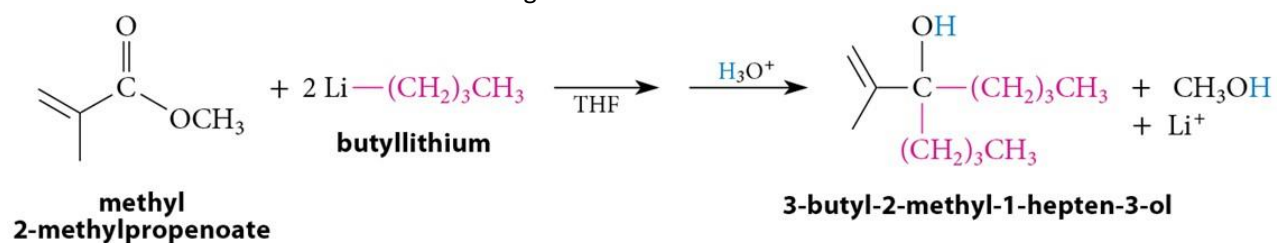


Mechanism questions with arrow pushing: Make sure to push the arrows to the correct direction! Nucleophile should attack, but not be attacked. Electrophile should be attacked, but not attack.

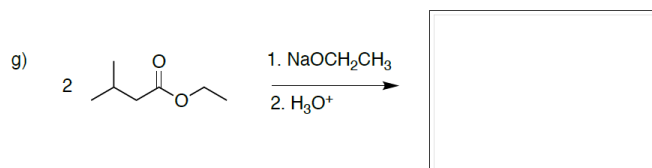
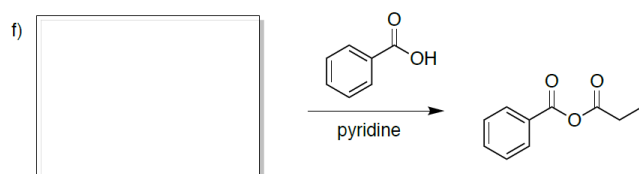
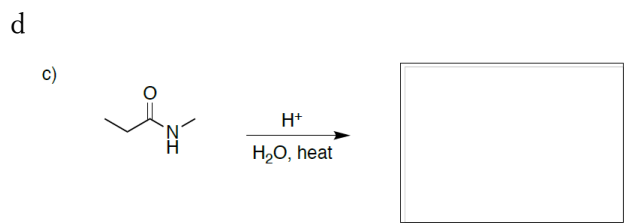
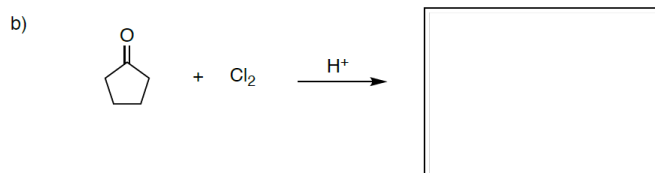
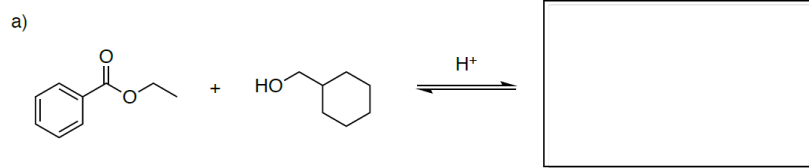
4. Provide a mechanism for the following reaction:



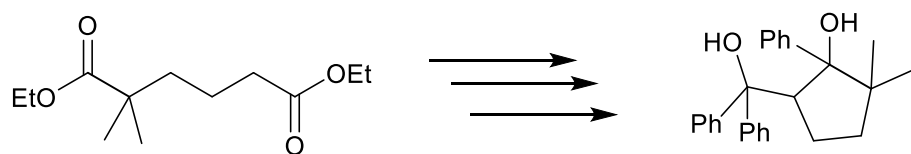
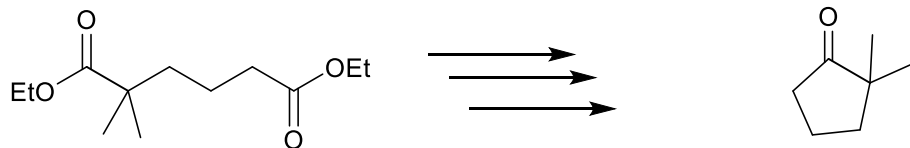
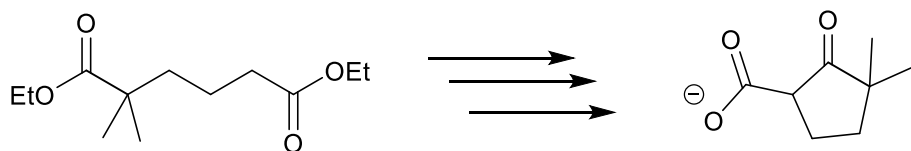
Provide a reaction mechanism of the following reaction.



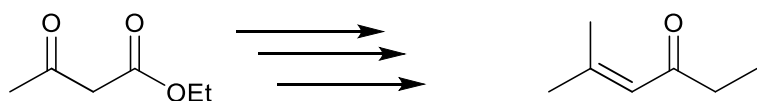
Topic list actually said that the mechanism of the reaction of esters with Grignard reagents (or alkyllithium) don't have to be memorized. So, you don't need to work on this one.



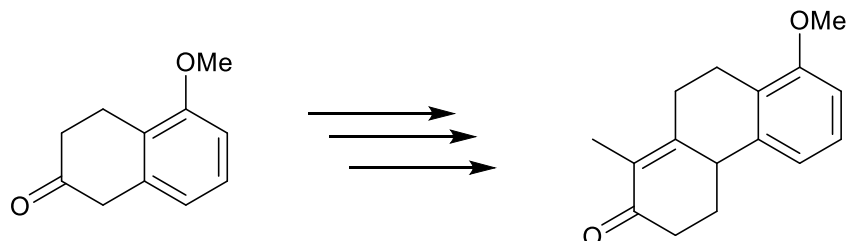
Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



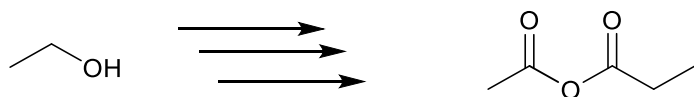
Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.

